ME Grad Assignment 4.1

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March 6, 2026

Question 1

A python code was made to plot the data in semi-log-x graphic domain, iteratively find the logarithmic region, and fit the following logorathmic form to the appropriate region:

$$\eta = A - B\log(-I), \quad where \quad A = 2.303\frac{\bar{R}T}{\alpha_c F}\log(I_0),\ B = 2.303\frac{\bar{R}T}{\alpha_c F}.$$

Once A and B are determined from the fit, the code solves for I_0 and α_c. The results are shown in Figure 1.

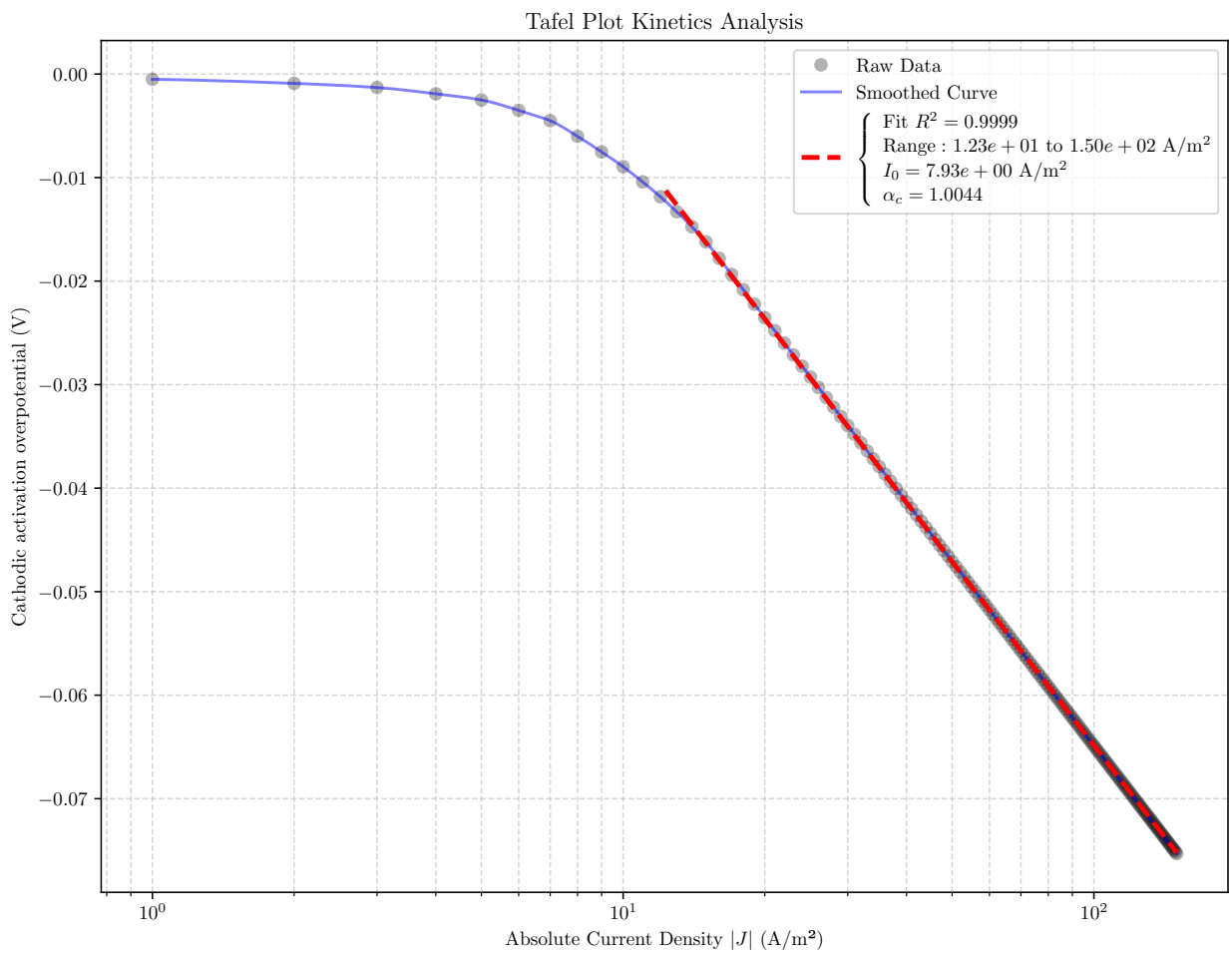

Figure 1: Tafel Slope with fitted regression and parameter outputs.

Question 2

$$\eta_{ohmic} = j \cdot \frac{L}{\sigma}$$

$$\eta_{ohmic} = 1\frac{A}{cm^2} \cdot \frac{0.01\ cm}{0.1\frac{S}{cm}}$$

$$\boxed{= 0.1\ V}$$

Question 3

$$ASR_{117} = \frac{L}{\sigma} = \frac{0.007\ in \cdot 2.54\frac{cm}{in}}{0.05\frac{S}{cm}} = 0.3556\ \frac{\Omega}{cm^2}$$

$$ASR_{HP} = \frac{L}{\sigma} = \frac{0.02\ mm \cdot 0.1\frac{cm}{mm}}{0.05\frac{S}{cm}} = 0.04\ \frac{\Omega}{cm^2}$$

$$\%_{decrease} = 100 \cdot \left(1 - \frac{Final}{Initial}\right) = 1 - \frac{0.04}{0.3556}\ \boxed{= 88.75\%}$$

$$\eta_{117} = J \cdot ASR = 1\frac{A}{cm^2} \cdot 0.3556\ \frac{\Omega}{cm^2} = 0.3556\ V$$

$$\eta_{HP} = J \cdot ASR = 1\frac{A}{cm^2} \cdot 0.04\ \frac{\Omega}{cm^2} = 0.04\ V$$

$$\eta_{117} - \eta_{HP} = 0.3556\ V - 0.04\ V\ \boxed{= 0.3156\ V \text{ decrease}}$$

Question 4

$$\dot{n} = \frac{i}{nF}, \quad where \quad n = 2 \text{ electron mole-eq's per mole } H_2 \quad and \quad F = 96485\ C/mol$$

$$\dot{n} = \frac{1000\ A}{2\ mol \cdot 96485\ C/mol}\ \boxed{= 0.005\ mol/s}$$

Code Availability

The code used to generate the plot and fit is available in my GitHub repository: https://github.com/Ben-JaminRees/Fuel-Cell-Course-Spring-2026.